

Niyanta Pandey

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SUMMARY

Analytics Professional with 4+ years of experience delivering insights across healthcare, logistics, and e-commerce. Proficient in SQL, Python, and BI tools to drive EDA, experimentation, and executive reporting. Skilled in A/B testing, KPI definition, data modeling, and building dashboards that support business decision-making.

EDUCATION

Yeshiva University – New York, NY

Master of Science in Data Analytics | CGPA: 3.77/4.0

Aug 2023 – May 2025

Mody University – Rajasthan, India

Bachelor of Technology in Electronics & Communication Engineering

Aug 2014 – May 2018

TECHNICAL SKILLS

Languages: Python, SQL (PL/SQL), Shell scripting

Databases & Warehousing: PostgreSQL, Snowflake, Redshift, MySQL, MongoDB, PySpark

Data Processing & Orchestration: Pandas, NumPy, dbt, Apache Airflow, AWS Glue

Visualization & BI: Power BI, Tableau, Looker Studio

Product Analytics & User Behavior: Mixpanel, FullStory

Cloud Platforms: AWS (S3, IAM, Redshift, Athena), Azure (ADF, Synapse)

Experimentation & Analytics: A/B testing frameworks, statistical analysis

Data Privacy & Compliance: HIPAA, GDPR, PII protection

GenAI & NLP: OpenAI (GPT), Hugging Face, LangChain

ML & Modeling: scikit-learn, XGBoost, CatBoost, LightGBM, TensorFlow

WORK EXPERIENCE

Redistri B.V (Now Quicargo)

Analytics Engineer

Sep 2021 – Sep 2023

- Engineered analytics workflows using dbt and Python, to transform freight event streams into actionable insights for delivery monitoring to reduce freight delays.
- Optimized PostgreSQL queries, boosting dashboard responsiveness and query performance by 60%.
- Automated data validation with pytest, ensuring data accuracy across multi-source freight datasets.
- Collaborated with product and operations teams to analyze third-party route APIs via A/B and multivariate testing, improving routing efficiency by 15%.
- Defined and tracked detailed logistics KPIs, integrating them into executive dashboards that supported strategic decision-making.
- Performed segmentation and cohort analysis to identify routing and load distribution patterns, enabling targeted route API improvements that enhanced load utilization.

Leapfrog Technology, Inc.

Data Analyst

Jan 2019 – Sep 2021

PROJECTS UNDERTAKEN

MinderaDx, USA (Dermal Intelligence Platform)

- Analyzed clinical trials, manufacturing, genomics, and commercial operations data to support research and reporting.
- Developed and maintained data warehouse in PostgreSQL using Python, PySpark, SQL, AWS Lambda, and AWS Glue; ingested data from MSSQL, SFTP, and other sources.
- Integrated phenotype and genotype data to support classification model development for drug recommendations.
- Created and managed data lake in Amazon Athena to enable data access and cleaning workflows.

Laudio, USA (Healthcare Workforce Analytics)

- Applied NLP (sentiment analysis, topic modeling) on patient feedback to extract themes; insights used by product and UX teams to boost app retention by 24%.

- Tracked user behavior, analyzed session flows and drop-offs, informing product optimizations to increase staff engagement.
- Automated reporting and alerts with Power Automate, cutting manual effort by 40%.
- Designed and managed Airflow workflows integrating EMR, nurse schedules, and behavioral data.
- Refactored Redshift schemas to star schema, optimizing queries and reducing report latency by 67%.
- Created Power BI dashboards for shift compliance, care engagement, and KPIs; used daily by 4+ hospital teams.
- Enforced HIPAA-compliant governance with PII masking, data lineage tracked in GitHub, and role-based access control.

Aspen Biosciences (Drug Discovery & Management Platform)

- Built ETL pipelines using Python, SQL, and AWS Glue to process 10TB+ of sequencing data from FASTQ/S3 and MSSQL sources.
- Improved Athena query performance by 40% via optimized partitioning and Glue Catalog configuration.
- Designed executive dashboards in Power BI to track lab throughput, error rates, and manufacturing KPIs.
- Used Excel for QC audit logs and cross-validation of bioinformatics metrics.

Returnalyze Inc, USA (Retail Returns & Sales Intelligence)

- Defined statistical rules for anomaly detection in return patterns, incorporating segmentation by vendor and product category to detect fraud and defects.
- Built Tableau dashboards showing return reasons, warehouse mismatch rates, and SKU-level churn; implemented retention analysis to identify drivers of repeat returns.
- Conducted basket analysis and cohort segmentation to profile high-risk customer groups and inform targeted marketing strategies.
- Assisted in data governance efforts to standardize metrics definitions and automate manual validation workflows.

Accenture

Associate Data Engineer

Jun 2018 – Dec 2018

- Developed RESTful APIs using Flask and OAuth2, supporting integration into Jenkins CI/CD pipelines for automated deployment and testing of data services.
- Created documentation of data flows and business rules using Visio and Confluence to support QA, analysts, and engineers in understanding ETL pipeline logic.
- Performed data validation and root cause analysis (RCA) on ETL failures using SQL and log inspection, helping resolve anomalies and improve pipeline reliability.

ACADEMIC PROJECTS

- **Canvas LMS AI Assistant:** Built a production Retrieval-Augmented Generation (RAG) system using GPT-4, LangChain, and Pinecone serving 500+ users. Improved answer accuracy from 45% to 87% and reduced hallucinations by 61% through advanced prompt engineering and retrieval optimization.
- **Spotify Playlist Prediction:** Developed a hybrid predictive model by combining XGBoost and K-Nearest Neighbors (KNN) to forecast users' next tracks in playlists, leveraging both audio feature embeddings and contextual playlist metadata. Evaluated model performance using precision@k and Mean Reciprocal Rank (MRR), demonstrating improved recommendation relevance over individual models.
- **EEG Signal Classification:** Designed and implemented a robust EEG signal processing pipeline leveraging Support Vector Machines (SVM) to classify distinct cognitive states. Preprocessed raw EEG data through bandpass filtering and artifact removal techniques to enhance signal quality. Achieved 85% classification accuracy, surpassing baseline models by approximately 2% through systematic cross-validation and careful hyperparameter optimization.

CERTIFICATIONS

- **AI Agents**, Hugging Face (2025)
- **Databricks Lakehouse Fundamentals** (2024)
- **AWS Academy Cloud Architect** (2024)
- **AWS Academy Data Engineer** (2024)